Supporting Information

Mendelian randomization and multi-omic evidence for regulatory peak–gene links in human immune cells

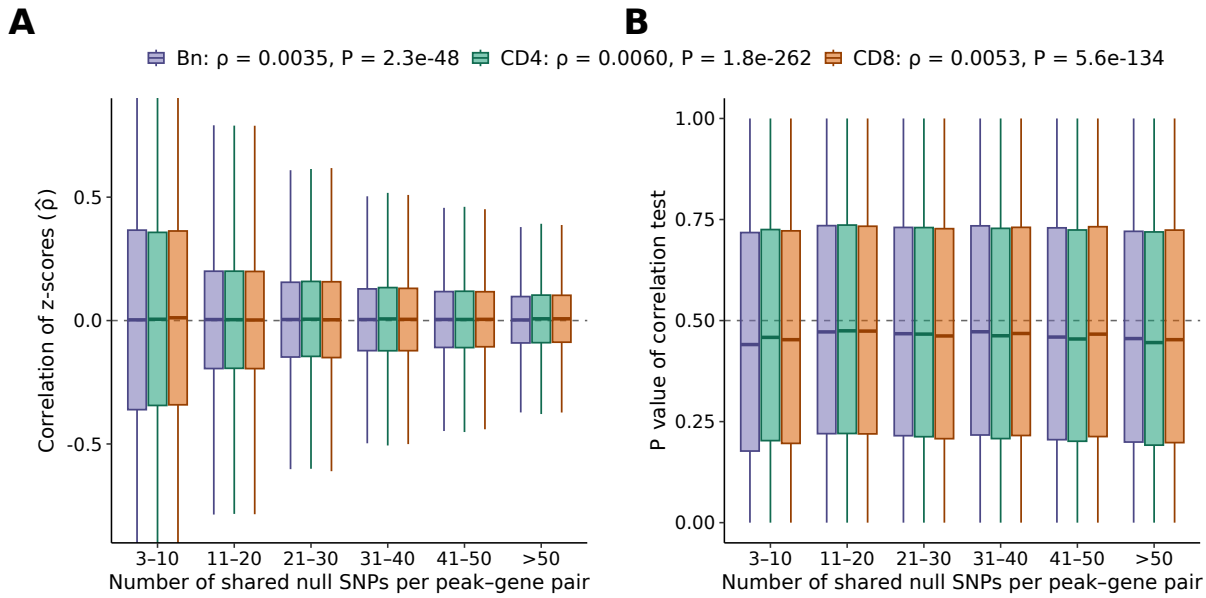
S1–S5 Figs and S1–S4 Tables accompany the main manuscript (`main.tex`).

Supporting figures

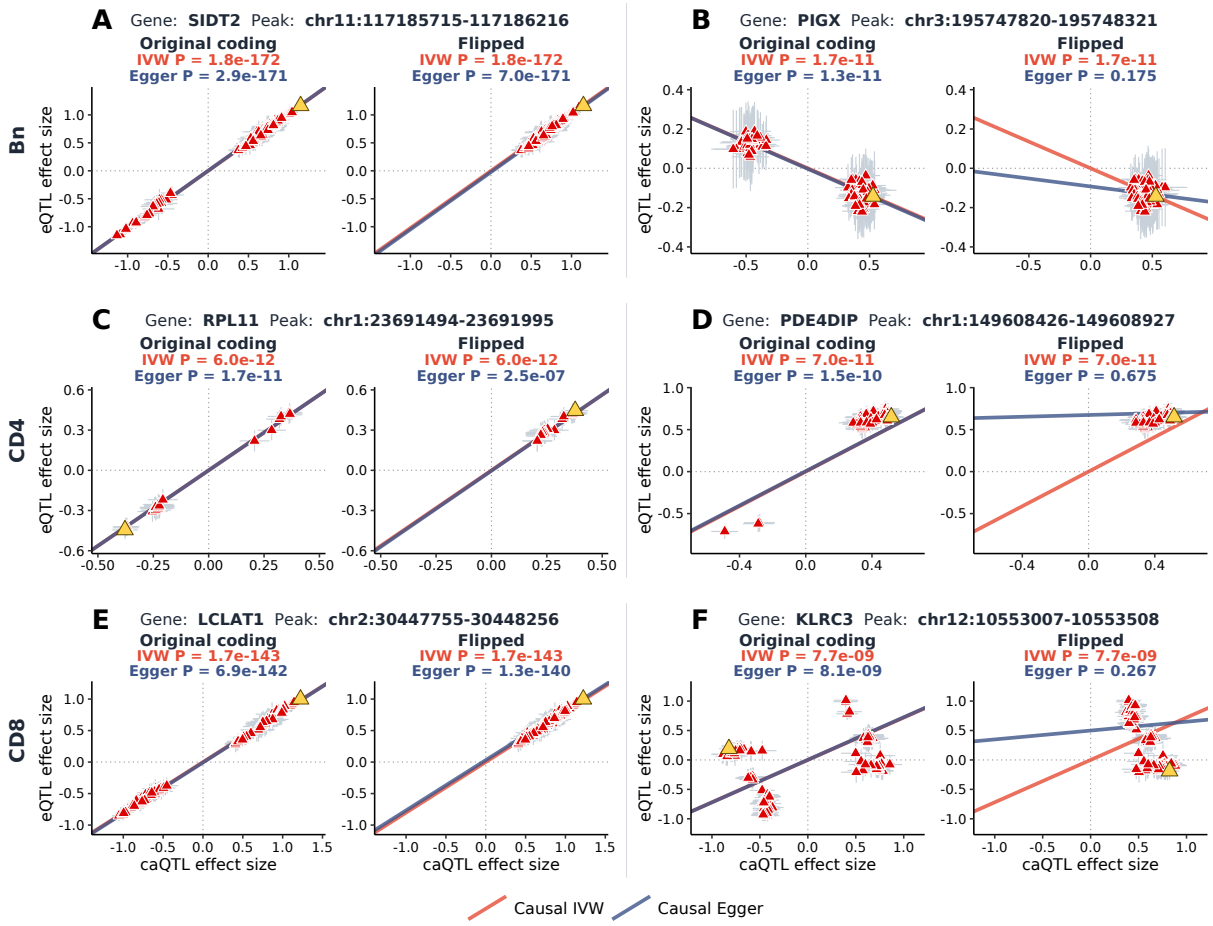
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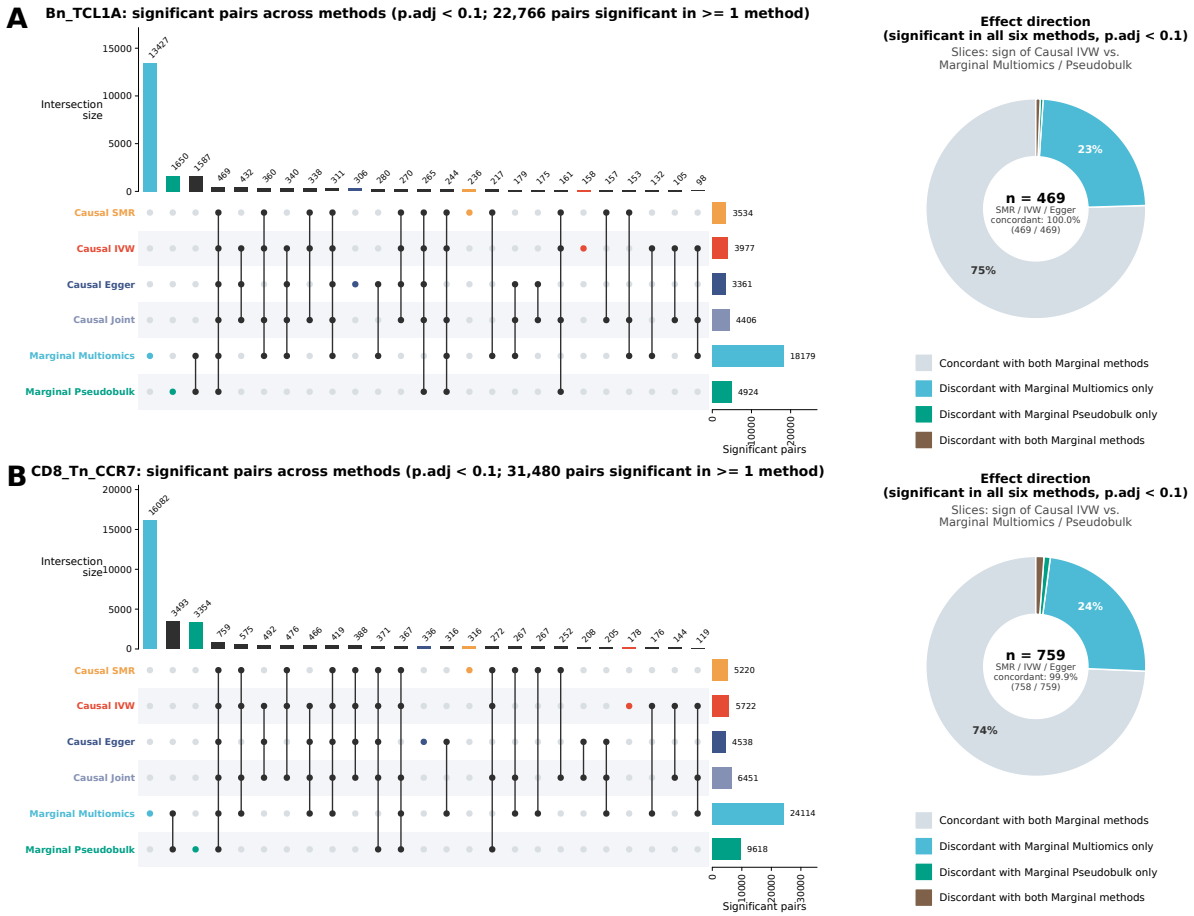
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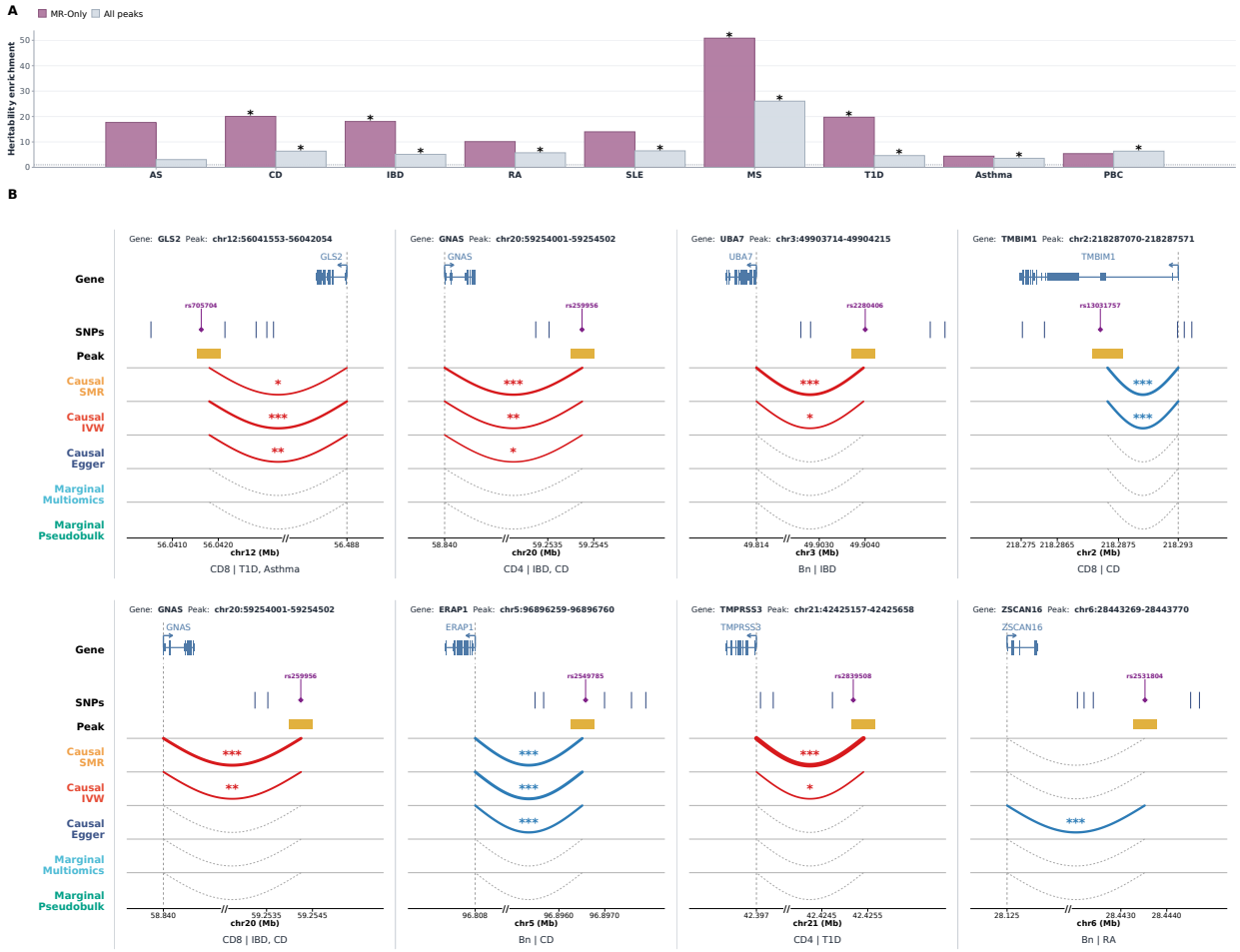
S1 Fig. Null-SNP diagnostics in Bn, CD4, and CD8 cells. **A,** Pair-level correlations of caQTL and eQTL z-scores, grouped by the number of shared pruned null SNPs; boxes show the three cell types side by side (Bn, purple; CD4, green; CD8, orange). The legend gives the global correlation ρ and its p-value for each cell type. **B,** Corresponding p-values of the pair-level correlation tests. The dashed lines mark zero correlation in A and a p-value of 0.5 in B, the values expected under the null. The analyses included 547,452 (Bn), 1,051,953 (CD4), and 686,829 (CD8) peak–gene pairs, based on 17,074,938, 33,263,695, and 21,537,822 SNP–pair observations, respectively. Global summaries pool SNP–pair observations, so a variant may occur in multiple pairs. This diagnostic does not itself establish MR calibration.



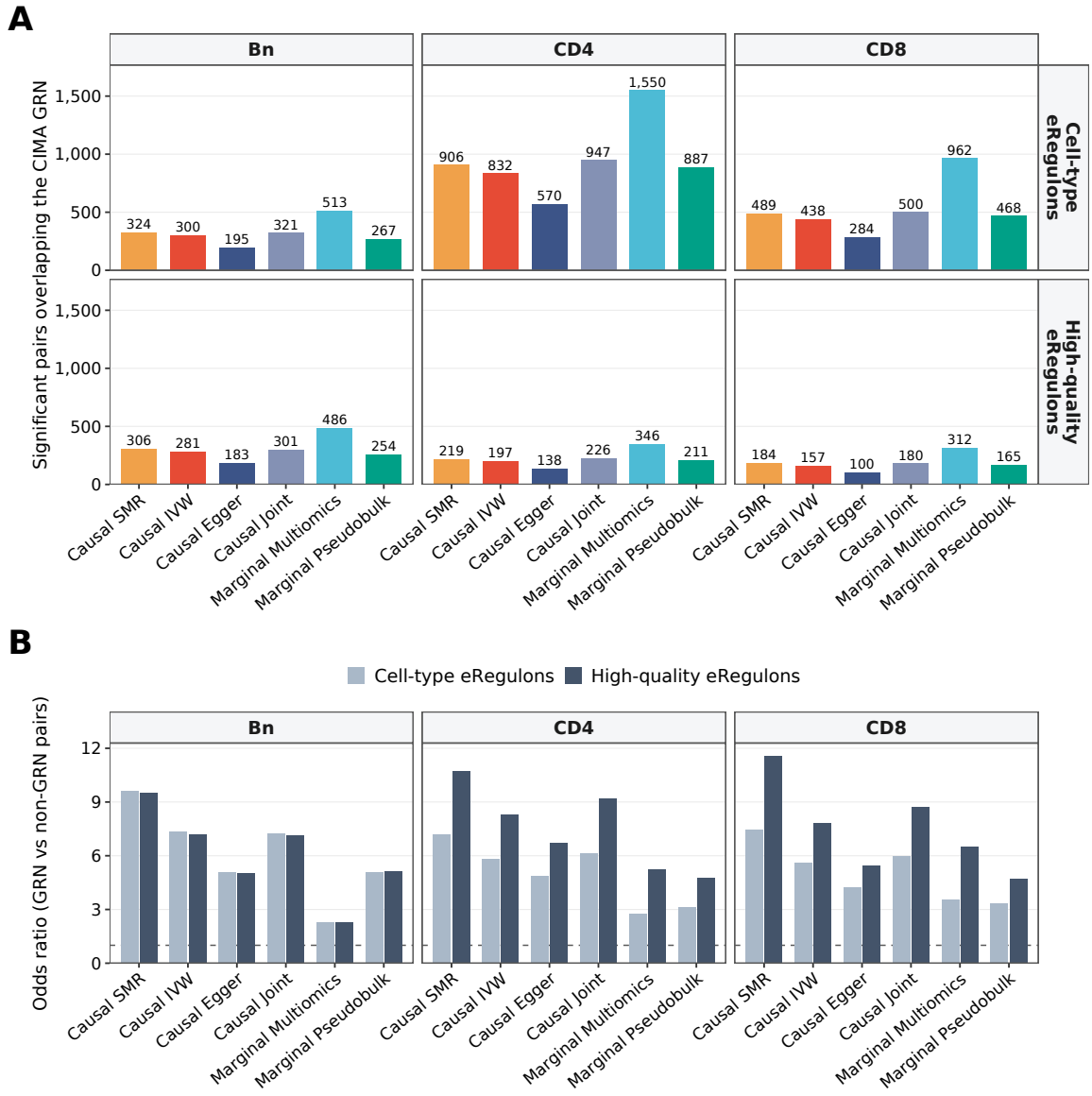
S2 Fig. MR-Egger estimates with and without allele orientation in Bn, CD4, and CD8 cells. Rows show Bn, CD4, and CD8 cells, and each lettered pair of plots is one peak–gene pair. In each row, the left pair of plots shows a pair whose estimates change little with orientation (A, SIDT2 in Bn, 194 SNPs; C, RPL11 in CD4, 25 SNPs; E, LCLAT1 in CD8, 195 SNPs) and the right pair shows a pair whose MR-Egger estimate changes markedly (B, PIGX in Bn, 136 SNPs; D, PDE4DIP in CD4, 74 SNPs; F, KLRC3 in CD8, 134 SNPs). For each pair, the left plot uses the original allele coding and the right plot orients the pair so that the caQTL effect is positive, as in the MR analysis; both plots share axis ranges. Red and blue lines are the IVW and MR-Egger fits, and the P values above each plot are the corresponding IVW and MR-Egger P values. Gold triangles mark the SMR lead SNP and gray bars are 95% confidence intervals. IVW is unchanged by orientation.



S3 Fig. Overlap and effect directions of significant links in Bn and CD8 cells. **A,** Bn; **B,** CD8. In each, left: the 25 largest distinct significance combinations across six methods at within-gene BH-adjusted $q < 0.1$; right: IVW direction relative to the two marginal correlation signs among links significant in all six methods; the center reports SMR/IVW/Egger agreement. "Causal Joint" denotes CCT. The corresponding CD4 comparison is shown in Fig 1B–C.



S4 Fig. Additional GWAS-overlapping regulatory links. **A,** The combined S-LDSC summary repeated from Fig 4A. **B,** Eight further loci in two rows. First row: GLS2 in CD8, GNAS in CD4, UBA7 in Bn, and TMBIM1 in CD8, with displayed GWAS overlaps T1D/asthma, IBD/Crohn's disease, IBD, and Crohn's disease, respectively. Second row: GNAS in CD8, ERAP1 in Bn, TMPRSS3 in CD4, and ZSCAN16 in Bn, with displayed GWAS overlaps IBD/Crohn's disease, Crohn's disease, T1D, and rheumatoid arthritis, respectively. Panel B follows the conventions of Fig 4B: each locus is titled with the gene and the hg38 peak coordinates and labeled below with the cell type and GWAS trait(s); purple diamonds mark GWAS variants; red/blue links indicate positive/negative effects, with line width increasing with $-\log_{10}$ adjusted p; gray dashed links are nonsignificant; link stars denote adjusted $p < 0.05$, 0.01, and 0.001; the axis break (//) compresses genomic distance.



S5 Fig. Overlap of significant links with the CIMA regulatory network. **A,** Number of links significant for each method (within-gene BH-adjusted $p < 0.05$, a stricter threshold than the $q < 0.1$ used elsewhere) that are also present in the CIMA SCENIC+ regulatory network. Rows show eRegulons active in the cell type (cell-type eRegulons) and the subset of these that belong to the high-quality eRegulons of the corresponding lineage (high-quality eRegulons); columns show Bn, CD4, and CD8 cells. **B,** Odds ratio for a significant link being present in the network, compared with links not in the network (2×2 table; all tested pairs in the cell type as background). Fisher's exact test $P < 10^{-30}$ for every bar, and the dashed line marks an odds ratio of 1. The cell-type eRegulons contained 864, 2,520, and 1,364 tested pairs in Bn, CD4, and CD8 cells, and the high-quality eRegulons 817, 458, and 382. "Causal Joint" denotes CCT.

S1 Table. Pooled hypergeometric enrichment p-values across regulatory references.

Cell type	Method	EnhancerAtlas	FANTOM5	4DGenome	GTEx	HiChIP	PCHiC
Bn	Causal SMR	3.7×10^{-136}	3.9×10^{-100}	2.5×10^{-58}	2.3×10^{-82}	1.0×10^{-416}	3.4×10^{-58}
	Causal IVW	1.2×10^{-108}	1.5×10^{-88}	3.1×10^{-52}	1.6×10^{-71}	2.7×10^{-365}	2.6×10^{-46}
	Causal Egger	6.0×10^{-65}	8.5×10^{-68}	5.6×10^{-40}	4.2×10^{-38}	9.0×10^{-286}	1.1×10^{-31}
	Causal Joint	3.7×10^{-130}	1.1×10^{-99}	2.3×10^{-54}	4.2×10^{-77}	5.3×10^{-427}	9.3×10^{-57}
	Marginal Multiomics	6.1×10^{-138}	1.3×10^{-162}	5.1×10^{-61}	2.5×10^{-61}	2.1×10^{-879}	2.9×10^{-75}
	Marginal Pseudobulk	1.6×10^{-80}	2.6×10^{-94}	2.4×10^{-44}	8.4×10^{-55}	8.0×10^{-404}	4.0×10^{-54}
CD4	Causal SMR	3.5×10^{-302}	6.5×10^{-107}	9.4×10^{-50}	2.6×10^{-209}	1.3×10^{-911}	2.0×10^{-140}
	Causal IVW	1.2×10^{-221}	3.0×10^{-94}	3.2×10^{-47}	3.6×10^{-151}	5.0×10^{-890}	2.2×10^{-154}
	Causal Egger	2.1×10^{-109}	4.9×10^{-47}	5.7×10^{-33}	9.2×10^{-98}	3.0×10^{-584}	1.9×10^{-92}
	Causal Joint	6.0×10^{-249}	6.2×10^{-109}	1.1×10^{-55}	8.2×10^{-195}	4.7×10^{-981}	5.6×10^{-160}
	Marginal Multiomics	1.7×10^{-200}	3.2×10^{-123}	6.1×10^{-40}	2.3×10^{-114}	6.2×10^{-1875}	1.3×10^{-191}
	Marginal Pseudobulk	1.3×10^{-137}	4.7×10^{-92}	4.1×10^{-37}	5.3×10^{-97}	1.6×10^{-1172}	1.4×10^{-158}
CD8	Causal SMR	2.1×10^{-204}	1.8×10^{-72}	2.4×10^{-38}	1.7×10^{-104}	1.6×10^{-614}	8.6×10^{-103}
	Causal IVW	3.1×10^{-156}	1.5×10^{-60}	1.7×10^{-34}	8.7×10^{-100}	1.1×10^{-616}	6.9×10^{-83}
	Causal Egger	3.4×10^{-104}	3.1×10^{-38}	1.2×10^{-28}	6.4×10^{-55}	1.2×10^{-446}	7.1×10^{-48}
	Causal Joint	8.8×10^{-189}	1.4×10^{-76}	3.9×10^{-41}	1.5×10^{-119}	4.6×10^{-687}	4.2×10^{-95}
	Marginal Multiomics	1.7×10^{-150}	3.1×10^{-111}	2.1×10^{-34}	2.9×10^{-75}	7.4×10^{-1466}	4.1×10^{-130}
	Marginal Pseudobulk	1.6×10^{-104}	1.4×10^{-77}	3.6×10^{-17}	9.5×10^{-57}	1.0×10^{-824}	1.7×10^{-77}

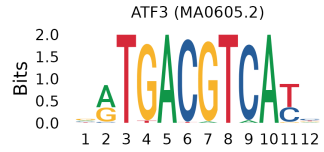
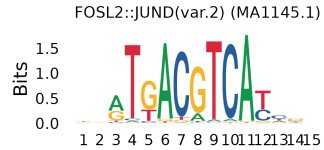
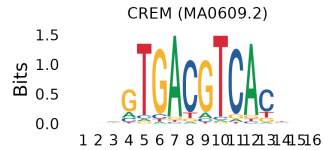
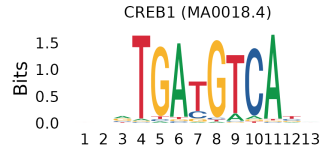
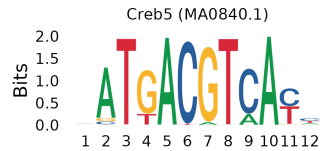
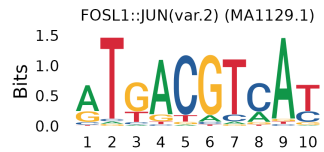
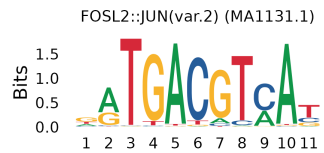
For each cell type, method, and reference, an upper-tail hypergeometric test was applied to counts pooled over eligible genes. For each gene, N is the number of background candidate peaks, K the number overlapping the reference, n_{sig} the number of selected significant peaks, and k the selected peaks overlapping the reference. Eligibility requires $N > 0$, $K > 0$, $n_{\text{sig}} > 0$, and $N > K$. The test uses $\sum N, \sum K, \sum n_{\text{sig}}, \sum k$. Entries are hypergeometric p-values, with no additional adjustment across the displayed comparisons specified by the source workbook. Bn, naive B cells; CD4/CD8, naive CD4/CD8 T cells. Causal Joint is the Cauchy combination (CCT) of SMR, IVW, and MR-Egger; Marginal Multiomics denotes the WNN metacell correlation and Marginal Pseudobulk the individual-level pseudobulk correlation.

S2 Table. Pooled regulatory-reference fold enrichment across cell types and methods.

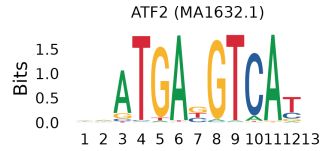
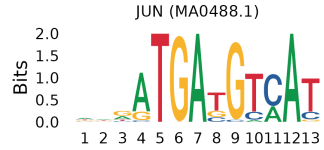
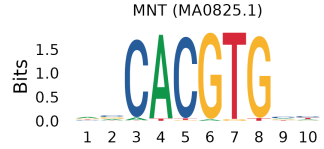
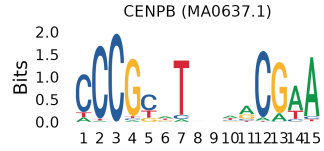
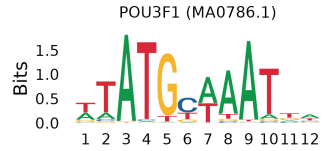
Cell type	Method	EnhancerAtlas	FANTOM5	4DGenome	GTEx	HiChIP	PCHiC
Bn	Causal SMR	**3.59**	**6.18**	**5.00**	**2.96**	**4.46**	**2.70**
	Causal IVW	3.10	5.65	4.54	2.71	4.02	2.46
	Causal Egger	2.69	5.59	4.39	2.34	3.89	2.32
	Causal Joint	3.20	5.67	4.42	2.70	4.12	2.56
	Marginal Multiomics	1.98	3.56	2.61	1.70	3.08	1.84
	Marginal Pseudobulk	2.49	5.03	3.76	2.34	3.76	2.38
CD4	Causal SMR	**3.56**	**4.45**	**3.30**	**3.01**	**4.65**	2.55
	Causal IVW	2.98	4.08	3.12	2.63	4.41	**2.61**
	Causal Egger	2.52	3.46	3.02	2.51	4.17	2.41
	Causal Joint	2.99	4.17	3.20	2.77	4.40	2.56
	Marginal Multiomics	1.84	2.49	1.85	1.68	3.35	1.85
	Marginal Pseudobulk	2.05	3.10	2.30	1.94	3.78	2.16
CD8	Causal SMR	**3.65**	**4.64**	**3.60**	**2.76**	**5.46**	**2.70**
	Causal IVW	3.13	4.19	3.36	2.69	5.26	2.52
	Causal Egger	2.90	3.84	3.48	2.40	5.08	2.29
	Causal Joint	3.21	4.41	3.46	2.75	5.22	2.53
	Marginal Multiomics	1.90	2.81	2.00	1.68	4.01	1.88
	Marginal Pseudobulk	2.21	3.60	2.11	1.96	4.62	2.07

Fold enrichment is $(\sum k/\sum n_{\text{sig}})/(\sum K/\sum N)$, using the eligible-gene counts defined in S1 Table. Values are rounded to two decimal places. Bold indicates the greatest unrounded enrichment within each cell type and reference column. Contributing genes and selected-set sizes can differ across methods; these descriptive values are not paired tests of method superiority. Bn, naive B cells; CD4/CD8, naive CD4/CD8 T cells. Causal Joint is the Cauchy combination (CCT) of SMR, IVW, and MR-Egger; Marginal Multiomics denotes the WNN metacell correlation and Marginal Pseudobulk the individual-level pseudobulk correlation.

S3 Table. MR-specific transcription factor motifs in peaks supported by Mendelian randomization.

Motif logo	p.adj						Brief interpretation	Reference (PMID)
	IVW	MR-Egger	SMR	CCT	WNN	Pseudo-bulk		
Bn (10 motifs)								
	0.041	**1.5e−4**	0.157	0.059	0.072	0.056	ATF3 drives PDL1 on mouse marginal zone B cells and restrains TLR-induced IL-6/IL-12b in macrophages; T-cell ATF3 protects against colitis.	28414328 16688168 30862736
	0.057	**0.001**	0.137	0.056	0.088	0.059	Fra-2 (FOSL2) regulates mouse B-cell proliferation and differentiation; JunD is induced by BCR/CD40 signals and elevated in SLE patient B cells.	28566276 7534795 39220919
	0.118	**0.003**	0.604	0.229	0.202	0.232	CREMalpha is elevated in SLE patient T cells and epigenetically controls IL2 and IL17A; CREM also restrains CAR/IL-15-driven NK-cell function.	23019580 40468083
	0.128	**0.006**	0.408	0.140	0.111	0.367	In mature B cells, BCR-induced CREB-1 phosphorylation drives junB and bcl-2 transcription through CRE elements; T-cell CREB promotes Th17-mediated autoimmunity.	8683108 8816467 29050947
	0.353	**0.006**	0.491	0.393	0.231	0.159	Creb5 was identified as a shared regulon in lung CD11b- dendritic cells during mouse asthma exacerbation, and Creb5 siRNA attenuated exacerbation.	36038770
	0.511	**0.006**	0.392	0.250	0.152	0.230	Fra-1 (FOSL1) restrains plasma-cell differentiation and rheumatoid-factor production in mouse B cells; c-Jun-deficient B cells proliferate and secrete Ig normally.	25288397 30779858 7889400
	0.302	**0.007**	0.468	0.235	0.112	0.091	Fra-2 (FOSL2) promotes mouse B-cell proliferation and pre-B differentiation via Foxo1/Irf4; c-Jun-deficient B cells proliferate and secrete Ig normally.	28566276 34066181 7889400
	0.314	**0.009**	0.392	0.292	0.122	0.189	ATF7 represses innate-immune genes in macrophages via G9a and mediates LPS-induced innate memory; ATF7 is reduced in ulcerative colitis mucosa.	26322480 40586859

Continued on next page

Motif logo	p.adj IVW	p.adj MR-Egger	p.adj SMR	p.adj CCT	p.adj WNN	p.adj Pseudo-bulk	Brief interpretation	Reference (PMID)
	0.113	**0.009**	0.198	0.255	0.088	0.261	ATF-2 and Jun bind the composite kappa3/CRE element of the TNF promoter in A20 B cells, a gene rapidly induced after BCR or CD40 stimulation.	8816436
	0.249	**0.010**	0.442	0.195	0.069	0.059	Jun binds the TNF promoter with ATF-2 in A20 B cells; c-Jun-deficient mouse B cells activate normally; a JUN SNP associates with SLE in Chinese.	8816436 7889400 22489574
CD4 (2 motifs)								
	0.176	**0.003**	0.051	0.032	0.077	0.113	Mnt deletion in mouse T cells caused Th1-skewed CD4 hyperproliferation, intestinal inflammation and lymphoma, linking MNT to T-cell homeostasis.	16507988 23150551
	0.035	**0.004**	0.172	0.033	0.329	0.156	CENP-B binds the CENP-B box in centromeric alpha-satellite DNA and is a major anti-centromere autoantigen, including in primary biliary cholangitis.	1730770 8527711 25281565
CD8 (1 motifs)								
	0.415	**0.006**	0.591	0.140	0.499	0.232	POU3F1 (Oct-6) is interferon-inducible in mouse macrophages and was among top predicted TFs for chromatin changes in SLE CD4+ T cells.	20687925 36738683

Motif enrichment was tested for peaks selected by each method (within-gene BH-adjusted $q < 0.1$) against a background matched on GC content and accessibility (Signac, JASPAR2020 vertebrate CORE motifs). The table lists MR-specific motifs: p.adj < 0.01 and fold enrichment > 1 in at least one of IVW, MR-Egger, and SMR (bold), with no significant enrichment (p.adj ≥ 0.05) in WNN or Pseudobulk, and a clear literature-based interpretation. p.adj is the adjusted motif-enrichment p-value reported by Signac for each method. Reference (PMID) lists the PubMed identifiers of the papers on which each interpretation is based. For each motif, the literature on the corresponding transcription factor(s) was searched in PubMed for functions in the same or a related immune cell type, or in immune-mediated disease, and every PMID was checked against its PubMed record (title, year, and abstract). Pseudobulk, pseudobulk correlation; WNN, WNN metacell correlation; CCT, Cauchy combination of SMR, IVW, and MR-Egger; Bn, naive B cells; CD4/CD8, naive CD4/CD8 T cells. Rows are ordered as in the source table.

S4 Table. Data overview for the three cell types analyzed.

	Bn	CD4	CD8
Single-cell data used for the pseudobulk correlation (after quality control)			
Individuals (passing RNA and ATAC cell-count filters)	335	372	346
scRNA-seq cells	160,577	916,643	460,439
scATAC-seq cells	123,580	360,343	210,077
scRNA-seq cells per individual, median (IQR)	436 (315–619)	2,336 (1,655–3,150)	1,200 (763–1,760)
scATAC-seq cells per individual, median (IQR)	348 (236–464)	898 (618–1,278)	510 (320–845)
Genes in the pseudobulk RNA matrix	36,326	36,326	36,326
Peaks in the pseudobulk ATAC matrix	338,036	338,036	338,036
cis-QTL summary statistics			
eQTL sample size	405	406	405
caQTL sample size	385	385	384
Peaks with at least one caQTL at $P < 10^{-8}$	3,449	6,110	3,847
Candidate peak–gene pairs and the Mendelian randomization analysis set			
Candidate gene–peak pairs (All_GenePeak, TSS–peak distance < 1 Mb)	3,814,826	5,146,441	4,505,251
With caQTL and eQTL variants in common (at least 3 shared variants)	46,190	109,961	62,136
Gene and peak present in both pseudobulk and WNN data			
After removing pairs with a missing p-value in any base method	40,153	85,782	51,123
Final analysis set (IVW residual variance ≤ 10)	40,087	85,527	50,931
Genes / peaks in the final analysis set	8,149 / 2,951	10,701 / 5,338	9,427 / 3,327
Variants (instruments) per pair in the IVW fit, median (IQR)	22 (10–40)	22 (10–43)	22 (9–41)
Samples for the marginal correlation methods			
Pseudobulk correlation: individuals	335	372	346
WNN correlation: metacells (external PBMC multiome)	80	80	80

Bn, naive B cells (CIMA L4 type Bn_TCL1A); CD4, naive CD4 T cells (CD4_Tn_CCR7); CD8, naive CD8 T cells (CD8_Tn_CCR7). The Chinese Immune Multi-Omics Atlas (CIMA; Yin et al., Science 2026; OMIX009786) comprises scRNA-seq and scATAC-seq data from 428 Chinese adults annotated into 73 L4 immune cell types. The cis-QTL were mapped within 1 Mb of the gene transcription start site or the peak midpoint, using genotypes of the whole cohort (5,051,397 variants with minor allele frequency > 0.1); the numbers in the table are specific to the three cell types analyzed here. Three sample definitions are used: the QTL sample sizes (summary statistics used by the Mendelian randomization methods), the individuals in the pseudobulk matrices (those passing the RNA and ATAC cell-count filters; pseudobulk correlation), and 80 metacells of an external PBMC multiome data set (WNN correlation). The caQTL and eQTL data come from the same cohort, so the two QTL data sets overlap in samples. Peaks with at least one caQTL at $P < 10^{-8}$ are counted from the significant caQTL files used to select instruments; this is a nominal threshold, not the false-discovery-rate-defined caPeak count of CIMA. Candidate pairs are All_GenePeak pairs with at least 3 variants shared between the clumped caQTL and the eQTL data, whose gene and peak are present in both the pseudobulk and WNN data. Pairs with a missing p-value in any base method (IVW, MR-Egger, SMR, WNN, pseudobulk) or with an IVW residual variance above 10 were removed. IQR, interquartile range.